

AI Meets Geometry

Lecture 1: AI Overview

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1 What is AI?

2 AI Subfields

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

3 Summary

What is AI?

AI $\implies$ Methods of Computational Statistics

- AI methods can: *approximate* functions, *recognise* patterns, *search* solution spaces.
 - Developed through python libraries such as PyTorch, TensorFlow, and JAX.
 - AI is optimal for mathematics since data is *cheap* to sample at *scale*, and *noiseless*.
 - Most modern hype surrounds LLM Agents, large AI models trained on all human knowledge and to use tools themselves.
- [These models are already assisting in mathematical formalisation, not our focus here.]

Machine Learning Subfields

- 1) **Supervised** learning: fit a map $\{\text{inputs}\} \mapsto \{\text{outputs}\}$ from labelled examples; ...tasks include classification and regression.
- 2) **Unsupervised** learning: analyse unlabelled data to find structure; ...typical goals are clustering, anomaly detection, and dimensional reduction.
- 3) **Reinforcement** learning: learn a sequence of actions that maximises long-term ...reward in a state space; this is useful for search and control.

Supervised Learning: Training

Supervised Training

- Start with labelled pairs (x_i, y_i) , for example coefficients of an equation and a target invariant.
- Choose an architecture f_θ with many trainable parameters θ .
- Define a loss $\mathcal{L}(f_\theta(x), y)$ measuring prediction error.
- Use an optimiser, such as SGD or Adam, to stochastically update θ on batches of data.
- Check generalisation on held-out data to assess performance, then analyse and interpret.

Standard Loss Functions

- Loss choice depends on the task, the architecture, and desired solution.
 - Classification $\implies \mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i,c=1}^{N,C} y_{i,c} \log \hat{p}_{i,c}$ (cross-entropy)
 - Regression $\implies \mathcal{L}_n = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|^n$ ($n=2 \implies \text{MSE}$)
 - Regularisation $\implies \mathcal{L}_{R,n} = \lambda \sum_p |\theta_p|^n$ ($L^1 \& L^2$ penalties)
- The optimiser follows gradients, schematically $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$.

PINN Losses

- For PDE problems the loss can encode the equation itself rather than labelled outputs.
- Differential losses are computed with automatic differentiation.
 - Boundary or initial conditions become extra penalty terms.
 - Extra constraints can repel known trivial solutions or enforce symmetries.

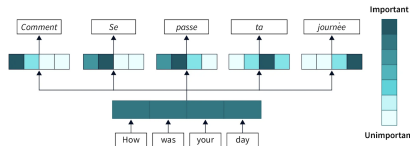
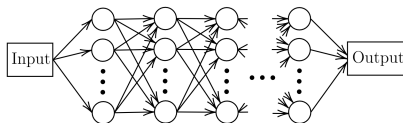
Supervised Learning: Architectures

Neural Networks

- A neural network is a highly-parameterised non-linear map built from affine maps and activation functions, layering linearity and non-linearity builds complexity and expressibility.

$$f_{NN} := W_{i_L, i_{L-1}} \cdot \alpha(\dots \alpha(W_{i_1, i_{input}} \cdot x_{input} + b_{i_1}) \dots) + b_{i_L}$$

- Universal approximation theorems say that sufficiently wide or deep networks can approximate continuous functions on compact sets; this is an existence result (not a training guarantee).



Transformers

- Transformers, the basis for LLMs, combine neural subarchitectures, especially NNs and...

$$\text{Attention}(Q, K, V) := \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

...notably successful on sequential data such as text, code, and tool instructions.

- A general empirical rule of AI is that architectures on tasks follow *scaling laws*:
...performance improves predictably as model size, data, and compute *increase*.

Supervised Learning: Interpretability

Supervised Models for Geometry

- Many geometric objects can be cheaply sampled to provide large datasets: *manifold points, moduli space points, tensors, immersion map coefficients...*
- Supervised models can approximate at varying complexity to build intuition.
- Trained models provide fast predictors for scanning large parameter spaces, ranking promising regions before exact analysis; and for extrapolation beyond known domains.

Interpretability Methods

- Interpretability is analysis of trained models to extract mathematical insight.
- Common methods include: feature importance, saliency, symbolic regression.
- Symmetry-aware architectures can test for, and uncover, invariances or equivariances present in the data, solution, or task.
- Trained models can predict verifiable counter-examples.
- Recently, PDE methods can now prove nearby existence and error bounds on AI approximations, leading to computer-assisted concrete proofs.

Unsupervised Learning: Analysis

Unsupervised Analysis

- **UN**supervised learning has no paired output to train against, their methods are effectively computationally-intensive statistical analysis.
- Data is of the form $\{x_i\}$, so could take any input or output dataset from supervised learning, looking to uncover mathematical structure.
- Typical questions: Are there natural families? Are there suitable lower-dimensional coordinates? Any unusual outliers?

Unsupervised Technique Classes

- Historically, two dominant classes of techniques:
 - Clustering**: Partition data into groups accordingly to similarity measures.
 - Dimensional-reduction**: Compress data, removing noise, focusing on dominant degrees of freedom.
- Unsupervised methods are often used throughout the pipeline on a AI-project: ...analysing the data before training, debugging training processes, analysing trained models or novel predictions after training.

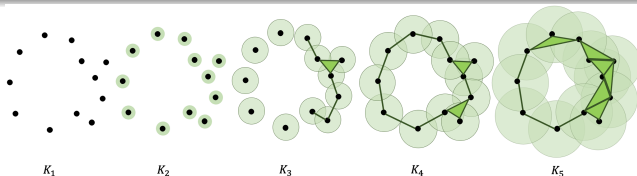
Unsupervised Learning: Clustering

Standard Methods

- The canonical clustering method is **K-means**:
...choose K random centres, assign each datum to its nearest centre, update the centres, reassign, iterate to convergence.
- This can be developed into **Kernel**, **Hierarchical**, **Spectral**, or **Density** methods,
...useful for irregular cluster shapes, or when outliers matter.

Topological Data Analysis (TDA)

- Most common method is *persistent homology*:
Treat datapoints as embedded 0-simplices, grow balls of increasing radii simultaneously, where intersections occur update the growing simplicial complex, track the homology of the complex for persistent features.



Unsupervised Learning: Dimensional Reduction

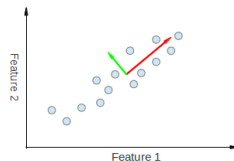
Standard Methods

- The canonical dimensional reduction method is **PCA**:
...linear projection onto directions of largest variance; simple and interpretable.
- NNs can be used via **Autoencoders**:
...output data equals the input, NN learns identity map through a lower-dimensional bottleneck, giving the latent representation.
- Other non-linear methods include **t-SNE** and **UMAP**:
...mainly visualisation tools, that minimise some difference measure with a lower-dimensional representation of the data distribution.

Unsupervised Methods for Geometry

- Identify redundant degrees of freedom: coordinates, submanifolds, invariants.
- Reveal effective coordinates on high-dimensional families of solutions, and symmetry-reduced descriptions.
- Visualise geometric objects in lower-dimensions.

PCA:



Reinforcement Learning: Training

RL Setup

- For RL problems you have a *state space* $\{s_t\}$ of potential solutions, and an *action space* $\{a_t\}$ of ways to perturb these solutions.
- The goal is to train a *policy* $\pi(a_t | s_t)$ which optimally selects the next action a_t given the current state s_t .
- Need a method of evaluating states, which dictates the *reward* r_t allocated per step in training.
- The training process looks to maximise long-term cumulative reward.

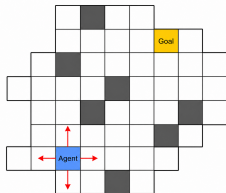
Discrete vs Continuous

- The state and action spaces may be discrete (pictured), continuous, or a mix.

Discrete spaces arise in combinatorics, symbolic manipulation, and graph search.

Continuous spaces arise in PDE parameter tuning, shape optimisation, and manifold search.

- RL useful when a good solution can be found from iterated perturbations from generic starting points.



Reinforcement Learning: Methods

Standard Methods

- The canonical RL method is **Deep-Q**:
...the state space may be continuous or discrete, the action space is discrete. **Value-based**, such that an NN predicts success of actions given a state, the policy is then a function of this (often argmax).
- These can develop into **Policy-based** methods:
...both state and action spaces may be continuous. The policy function is replaced directly with a NN, which is trained over trial runs.
- These can develop further to **Model-based** methods:
...the transition to the next state is now *probabilistically* dependent on the action, which may be approximated with a trained NN. The reward function may also be approximated by another NN; both used for simulation in training.

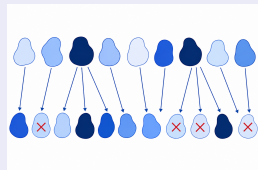
RL for Geometry

- RL methods do not require gradients, so useful where gradients are infeasible or expensive – especially for non-continuous or singular functions.
- Compatibility with discrete spaces means methods are well-suited for meshing and triangulation problems.

Reinforcement Learning: Genetic Algorithms

Genetic Algorithms

→ Predecessors to RL, genetic algorithms maintain a *population* of potential solutions, which are *mutated* by small perturbations (alike RL actions) to produce the next generation. These are evaluated and *fittest* survive.



- *Crossover* can be introduced where some interpolation function is used to make offspring for the next generation from pairs in the previous.
- These are gradient-free search methods, so they remain useful when the objective is noisy, discontinuous, combinatorial, or hard to differentiate.

Symbolic Regression

- This is the application of genetic algorithms to mathematical expressions.
- The population is a list of potential expressions, that can be mutated by a list of actions involving a basis of functions (e.g. $\{+, \times, \exp\}$ etc.).
- Expressions are evaluated on the data to give the fitness score.
- Can weight against complexity of expressions; directly interpretable outputs.

Summary

AI Overview

- AI methods are computational and statistical, they are useful for approximation, pattern finding, and search; there are *many* methods suited to different problems.
- AI roughly splits into supervised, unsupervised, & reinforcement methods.
 - Supervised** fits functions based on losses.
 - Unsupervised** analyses data and interprets trained models.
 - Reinforcement** perturbatively improves solutions, allowing discrete regimes.
- Geometry is a *phenomenally* optimal area for use of AI, since data is cheap to sample, noiseless, and is often infinite.
- AI methods are informative for uncovering relationships, building intuition, finding counter-examples, and more recently providing rigorous proofs.

Next Lectures

- Hands-on writing code in `python` for geometric applications:
Calabi-Yau metrics, Willmore minimisation, line arrangements.
 - ⇒ bring your laptops! ☺